# CSA1017 – SOFTWARE ENGINEERING

# INDIVIDUAL REFLECTION REPORT

## INTELLIGENT DISASTER EARLY WARNING AND EVACUATION MANAGEMENT SYSTEM

**NAME: DURGAM HITHESH**

**REGISTER NO.: 192472167**

**ROLE:** Shelter Management and Emergency Response Coordination

**DESIGN / DEVELOPMENT DECISION:** I worked on shelter capacity, occupancy, accessibility, food, water, medical support, sanitation, power, supplies, response-team status, assignments, and activity timelines. I supported a combined capacity-and-resource view because available physical space alone does not prove that a shelter is ready for use.

**CHALLENGE FACED AND SOLUTION:** The main challenge was representing changing shelter and response information consistently across different screens. I addressed this through structured shelter records, occupancy calculations, status badges, update forms, and a simple local-state approach suitable for the MVP.

**LEARNING GAINED:** I learned about data modelling, state management, resource allocation, shelter readiness, and inter-agency coordination. The project showed me how operational data must be organized so that authorities can quickly identify shortages, assign teams, and monitor progress during an emergency.

**SDG RELEVANCE:** The system relates to SDG 11 because shelter readiness and coordinated emergency response contribute to safer and more resilient communities.

**COURSE OUTCOME CONTRIBUTION:** My contribution supports the course outcome related to data management, system integration, and application workflow design through the shelter and response-coordination modules.

**POSSIBLE IMPROVEMENT:** A production system should include real inventory synchronization, audit logs, secure multi-agency access, and automated resource-shortage alerts.